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Analysis of Underwater Acoustic Communication System Using Equalization Technique for ISI Reduction

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Abstract

Underwater Acoustic Communication (UWAC) has been an active research area over a decade. Apart from regular communication inside the water, the applications include Remotely Operated Vehicles (ROV), Underwater Sensor Networks (UWSN), military and navigation applications, diving purposes, Autonomous Underwater Vehicles (AUV), underwater sports, etc. The major disadvantages of UWAC are frequency-dependent attenuation, short range of communication, very low bandwidth. In this paper, Modulation techniques such as BPSK, QPSK, and 4-QAM are used in UWAC in order to reach longer distances inside water. The fundamental key aspect of UWAC including significant multipath interference and strong ambient noise, are addressed. Equalization techniques are used at the receiver to overcome the Inter symbol Interference (ISI) caused due to multipath propagation. Zero Forcing (ZF) and Minimum Mean Square Error (MMSE) equalizers are used in this paper. However, ZF comes at the cost of increasing the noise; however; MMSE reduces all the noise levels in UWAC.

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Keywords: Underwater Acoustic Communication; ZF Equalizer; MMSE Equalizer; Modulation Techniques; ISI.

1. Introduction

Present-day research is focussing more on Underwater Communications because of its challenging tasks such as variations in undersea atmosphere and multipath fading [1]. The principal use of underwater acoustics (UWA) was

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initiated in 1490, by Leonardo Da Vinci, but this area of Communications came into actual existence after the tragic sinking of Titanic in the 20th century. The main applications of underwater communications are Remotely Operated Vehicles (ROV), Underwater Sensor Networks (UWSN), military and navigation applications, diving purposes, Autonomous Underwater Vehicles (AUV), underwater sports, etc. Amongst the various means of communication in water, Cables are considered to be more robust except that it is very tough to maintain undersea. Though EM waves are effective in power restricted region, it gets attenuated quickly inside water [2]. Considering all the above disadvantages, Optical underwater communication has drawn great attention for the last few years. This is considered to be economical and consumes low energy, which provides high data rates for short transmission distances. But the disadvantage of this means of communication is that the propagation of light underwater is affected by both absorption and scattering [3]. Also, the modulation bandwidth of LED for visible light communication is limited in underwater [4]. Irrespective of different signals that can be used to propagate in wireless media, sound (acoustic) signals produced by hydrophones are mainly used in underwater [5]. Because of the low absorption characteristic inside the water communicating through acoustic waves is preferable. This property of the acoustic waves inside the water enables the carrier to travel at the longer range at a slower rate. Typical frequencies associated with underwater are 10Hz to 1 MHz At a frequency lower than 10Hz, the propagation of sound is usually not possible while that higher than 1MHz are rarely used. A sound wave that is transmitted underwater has interchanging compressions and rarefactions which are detected by the receiver. The multipath problem in Underwater Acoustic Communications (UWAC) will lead to Inter Symbol Interference (ISI) and frequency selective fading [6]. The multipath property in UWAC causes time spread. If this time spread is larger than the symbol duration, ISI occurs. Digital modulation techniques are employed for better quality and efficient communication. Acceptable power, available bandwidth, and high noise immunity are the main advantages of digital communication. The same is used in UWAC in order to reach longer distances [7]. In this paper, the characteristics of Bit Error Rate (BER) for different modulation techniques are studied. The modulation techniques that are compared in this paper in the UWAC environment are BPSK, QPSK and 4-QAM. In this study, the authors have analyzed the performance of a UWAC system using different modulation techniques and the equalization techniques at the receiver. The transmission medium in communications systems introduces amplitude and phase distortion. These can be eliminated by a process known as Equalization. Any signal processing operation which minimizes ISI (in this case ISI is caused due to multipath in UWAC) is termed as Equalization. To decrease the interference mitigation, Zero Forcing (ZF) and Minimum Mean Square Error (MMSE) equalization techniques are introduced, which is the main focus of this study. This paper is structured as follows. The system model is explained in Section 2 considering the different modulations BPSK, QPSK, and 4-QAM in 2.1, 2.2, and 2.3, respectively. Section 3 deals with the Underwater Channel. Equalization techniques are illustrated in Section 4, details of ZF and MMSE in 4.1 and 4.2, respectively. Simulation results are depicted in Section 5, followed by the conclusions and future scope in Section 6.

2. System Model

The process of systematically modifying a well-known deterministic signal, called the carrier, in accordance with the information signal is called modulation. This process needs to be reversible, so that the receiver may be able to recover the information signal, via a process called the demodulation. This two-step process of modulation-demodulation is necessary when transmission of the signal directly in its original form is not possible through the channel, thus necessitating the translation of the information to another frequency band that suits the channel being used. The following Subsections help us to understand the modulation techniques used in this paper in a better way.

2.1. Binary Phase Shift Keying (BPSK)

BPSK is the basic modulation scheme which is considered to be the elementary form of Phase Shift Keying (PSK). In this modulation scheme, for every variation in the binary state carrier sine wave is shifted by 180°. As the phase alterations occur at the zero-crossing points, BPSK is considered to be coherent. Here the phase of the carrier

is controlled by the individual bits [8]. The binary bits 1 and 0 in this modulation scheme are represented by $+\sqrt{E_b}$ and $-\sqrt{E_b}$.

The modulator and the demodulator of BPSK modulation are as shown in Figure 1. A BPSK signal is described by the following equations

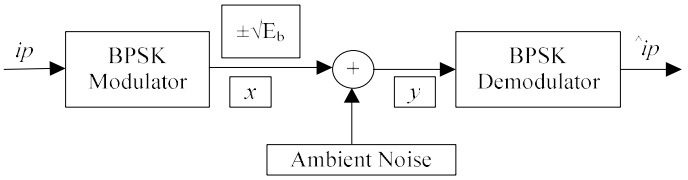


Fig 1. Block Diagram of BPSK

$$s_1(t) = \sqrt{\left(\frac{2E_b}{T_b}\right)}\cos(\omega_c t + \theta)(\theta = o) \tag{1}$$

$$s_2(t) = \sqrt{\left(\frac{2E_b}{T_b}\right)}\cos(\omega_c t + \theta + \pi) = -s_1(t) \tag{2}$$

T_b is the bit duration, E_b is the bit energy, ω_c represents angular frequency which is given by $\omega_c = 2\pi f_c$ and the initial phase θ that can be considered as zero. The received vector r is the vector sum of a signal point given by

$$r(t) = s(t) + n \tag{3}$$

Here n is the ambient noise explained in Section 3. According to Maximum Likelihood function, probability of BER is expressed as

$$f_r(\bar{r}/s_2(t)) = f_r(\bar{r}/'0') = \frac{1}{\sqrt{\pi N}}exp\left[-\frac{1}{N}(r - s_{21})^2\right] \tag{4}$$

$$= \frac{1}{\sqrt{\pi N}}exp\left\{\left[\frac{-\left[r-(-\sqrt{E_b}\right]}{N}\right]\right\}^2 \tag{5}$$

$$= \frac{1}{\sqrt{\pi N}}exp\left[-\frac{1}{N}(r + E_b)^2\right] \tag{6}$$

The conditional probability that the receiver decides in favor of '1' while '0' was transmitted

$$= \int_0^\infty f_r(r/0)dr = P_e(0)$$

$$\therefore P_e(0) = \frac{1}{\sqrt{\pi N}}exp\left[-\frac{1}{N}(r + E_b)^2\right]dr \tag{7}$$

Now consider $\frac{1}{\sqrt{N}}\left(r + \sqrt{E_b}\right) = Z$ then the above equation becomes

$$P_e(0) = \frac{1}{\sqrt{\pi}}\int_{\sqrt{E_b/N}}^{\infty}\exp\left(-Z^2\right)dz \tag{8}$$

Multiply and divide by equation (8) by a factor 2

$$P_e(0) = \frac{1}{2}\frac{2}{\sqrt{\pi}}\int_{\sqrt{E_b/N}}^{\infty} e^{\left(-Z^2\right)}dz \tag{9}$$

$$= \frac{1}{2}erfc\left(\sqrt{\frac{E_b}{N}}\right) = Q\left(\sqrt{\frac{2E_b}{N}}\right) \tag{10}$$

In a similar fashion

$$P_e(1) = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{N}}\right) \tag{11}$$

As discussed earlier in BPSK modulation '0's and '1's are equally likely to occur, so the average BER is given as

$$BER = \frac{1}{2}P_e(0) + \frac{1}{2}P_e(1) = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{N}}\right) \tag{12}$$

For demodulating a BPSK signal, the frequency of the coherent detector and the carrier signal at the transmitter need to be maintained equally. This leads to more complex receivers. The data rate in bits per second and the symbol rate are generally the same in BPSK. This is because a single bit is carried by a single carrier in this scheme. Therefore BPSK is not bandwidth efficient modulation technique compared to other modulation types. But to achieve higher data rates in UWAC, bandwidth has to be increased [9]. To overcome this, Quadrature Phase Shift Keying (QPSK) multi-level modulation techniques came into the picture in digital communications. Here modulation levels greater than two are considered as the transmitter's input. Therefore, these techniques are considered to be more efficient in terms of bandwidth.

2.2. Quadrature Phase Shift Keying (QPSK)

There is a special case in PSK wherein M=4, and each symbol represents 2 bits. This case is known as QPSK which is given by [10]

$$s_i(t) = \sqrt{\frac{2E_b}{T_s}} \cos\left(2\pi\omega_c t + (2i-1)\frac{\pi}{4}\right) \qquad 0 \le t \le T_s \tag{13}$$

The above equation can be written as

$$s_i(t) = \sqrt{\frac{2E_b}{T_s}} cos\left[(2i-1)\pi/4\right] cos\omega_c t - \sqrt{\frac{2E_b}{T_s}} sin\left[(2i-1)\pi/4\right] sin\omega_c t \tag{14}$$

Here *i=1, 2, 3, 4.* T_s is the symbol duration. The received QPSK signal is given $r_i(t) = s_i(t) + n(t)$. The ambient noise *n(t)* is explained in section 3. To reduce the complexity of computation, the probability of correct data that is to be transmitted in $s_4(t)$ is calculated.

$$P_{c_{s_4(t)}} = \int_0^\infty \frac{1}{\sqrt{\pi N}} exp\left[-\frac{\left(r_1 - \sqrt{\frac{E_b}{2}}\right)^2}{N}\right] dr_1 \int_0^\infty \frac{1}{\sqrt{\pi N}} exp\left[-\frac{\left(r_2 - \sqrt{\frac{E_b}{2}}\right)^2}{N}\right] dr_2 \tag{15}$$

As both the received vectors r_1 and r_2 are statistically independent, $\frac{r_j - \sqrt{\frac{E_b}{2}}}{\sqrt{N}}$ is replaced as *Z*. Here j =1, 2.

$$P_{c_{s_4(t)}} = \left[\frac{1}{\sqrt{\pi}} \int_{-\sqrt{\frac{E_b}{2N}}}^{\infty} e^{-Z^2}\right]^2 dz \tag{16}$$

We know that

$$\frac{1}{\sqrt{\pi}} \int_{-a}^{\infty} e^{-x^2} dx = 1 - \frac{1}{2} erfc(a) \tag{17}$$

On substituting

$$P_{c_{s_4(t)}} = \left[1 - \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{2N}}\right)\right]^2 \tag{18}$$

Since the probability of correct data being transmitted is calculated, now the probability of error is given as

$$P_{e_{s4(t)}} = 1 - P_{c_{s4(t)}} = erfc\left(-\sqrt{\frac{E_b}{2N}}\right) - \frac{1}{4}erfc^2\left(\sqrt{\frac{E_b}{2N}}\right) \tag{19}$$

Similarly $P_{e_{s_1}(t)} = P_{e_{s_2}(t)} = P_{e_{s_3}(t)} = P_{e_{s_4}(t)}$

Therefore the overall probability of error is given as

$$P_e = 4 \times \frac{1}{4}\left[erfc\left(\sqrt{\frac{E_b}{2N}}\right) - \frac{1}{4}erfc^2\left(\sqrt{\frac{E_b}{2N}}\right)\right] \tag{20}$$

In computing we have

$$P_e = \frac{1}{2}erfc\left(\sqrt{\frac{E_b}{N}}\right) \tag{21}$$

In QPSK modulation technique, two binary bits are represented by one complex symbol which requires more power to transmit. Therefore the QPSK modulation scheme is complicated in UWAC. Here the received data is decoded by estimation of phase states of the signal transmitted inside water. Hence moving further on with 4-QAM modulation whose major benefit is the bandwidth efficiency.

2.3. Quadrature Amplitude Modulation (QAM)

To represent binary data, both amplitude and phase variations are utilized in this scheme. As QPSK signal, 4-QAM also has both in-phase and Quadrature-phase (I and Q) signals. [11].They are as given below

$$s_I(t) = \sum_{k=-\infty}^{\infty}\sqrt{\frac{ET_Q}{T}}g_T(t-kT)cos\left(2I_k + 1 - \sqrt{M}\right) \tag{22}$$

$$s_Q(t) = \sum_{k=-\infty}^{\infty}\sqrt{\frac{ET_Q}{T}}g_T(t-kT)sin\left(2I_k + 1 - \sqrt{M}\right) \tag{23}$$

ET_Q is the minimum symbol of energy. Here M=4. If BER is calculated in the more general form then

$$P_{e,MQAM} \cong \frac{1}{\log_2\sqrt{M}}\left(1 + 2 + 4 + \ldots + 2^{\log_2\sqrt{M}-1}\right)P_Q \tag{24}$$

$$= \frac{2}{\log_2 M}(\sqrt{M} - 1)P_Q \tag{25}$$

$$P_Q = \frac{2}{\sqrt{M}}\sum_{i=1}^{\sqrt{M}/2} Q\left(\frac{(2i-1)d}{\sqrt{N/2}}\right) \tag{26}$$

Here P_Q is along the quadrature axis. For the above equation d can be calculated using the following relation

$$E_b \log_2\sqrt{M} = \frac{M-1}{3}d^2 \tag{27}$$

On substituting, the value of d in the BER equation,

$$P_{e,MQAM} \cong \frac{4}{\log_2 M}\left(1 - \frac{1}{\sqrt{M}}\right)\sum_{i=1}^{\sqrt{M}/2} Q\left((2i-1)\sqrt{\frac{3E_b\log_2 M}{(M-1)N}}\right) \tag{28}$$

Concentrating on the basic 4-QAM form now substituting M=4 in the above equations ,

$$P_{e,4QAM} = P_Q \tag{29}$$

Further evaluating P_Q is expressed as

$$P_Q = Q\sqrt{\frac{2E_b}{N}} \tag{30}$$

Expressing the above in error function terms

$$P_e = \frac{1}{2} erfc\left(\sqrt{\frac{E_b}{N}}\right) \tag{31}$$

QAM modulation is more inclined to the noise and bandwidth efficient inside the water as the states are closer. QAM receiver is more complex compared to the receivers of other modulation types. Considering the above-shown modulation schemes with its advantages and disadvantages, it is passed through an underwater channel which is explained in Section 3.

3. Underwater Channel

Because of the harmful effects of time and frequency spreading, communicating underwater has become the latest day challenge. Apart from the arrivals of multiple delay paths, attenuation and reflection loss causes the time spreading. Doppler shift is caused by the comparative motion between the transmitter and receiver, which impacts the frequency spreading [12]. The main difference between RF and UWAC channel is the propagation speed [13]. The above-explained modulation techniques increase the speed of acoustic waves inside the water. The characteristics that drive the complexity of the UWAC are the large delay spread; Doppler spread and the frequency dependence of propagation loss [14]. In water, the acoustic signals are generally affected by interference that is caused by reflection from the ocean surface and ocean bottom, as shown in Fig 2. As depicted along with the direct Line of Sight (LOS) propagation between the Tx and Rx, the signal propagates via multiple reflections from the ocean surface and the rough ocean bottom. Because of the rough scattering of these two surfaces, additional spreading occurs.

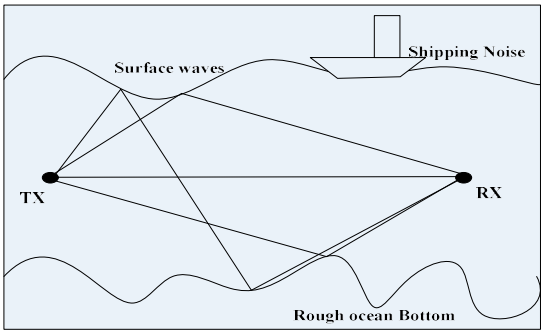


Fig 2. UWAC Channel with multipath

In modeling the underwater channel, the attenuation that is caused due to frequency absorption, ambient noise, and the loss due to the wave scattering at the surface and the bottom are considered. The two different losses that are considered in this paper are the spreading loss and absorption loss. As the wavefront moves away from the surface, the sound signal energy is covered by the same area, which causes the spreading loss. It is given by

$$SL(r) = k \times 10 \log(r) \tag{32}$$

Spreading loss is measured in dB. Here k is the spreading factor and r is the range in meters. In the case of unbounded spreading, k=2 while that in the case of bounded spreading k=1. For a short distance transmission (approx. 50m), spreading loss has a high impact. This also has a logarithmic relationship with the range 'r' [15]. As the sound waves propagate outwards underwater, the viscous friction and the iconic relaxation represents the energy loss in the form of heat. This is known as absorption loss, measured in dB given by

$$AL(r, f) = 10 \log(\alpha(f)) \times r \tag{33}$$

α is the absorption coefficient that is dependent on frequency.

Fisher and Simmons, Francois and Garrison proposed an empirical formula for α which varies with frequency, pressure (depth) and temperature [16]

$$\alpha(f, d, t) = \frac{A_1 f_1 f^2}{f_1^2 + f^2} + \frac{A_2 P_2 f_2 f_2^2}{f_1^2 + f^2} + A_3 P_3 f^2 \tag{34}$$

In this equation d is the depth, t is the temperature in °C. A represents the effects of temperature; P represents the ocean depth (Pressure) and f_1,f_2 represents the relaxation frequency. Attenuation/path loss that arises in UWAC channel for a distance l and signal frequency f is given by [17].

$$A(l,f) = A_0 l^k \alpha(f)^l \tag{35}$$

A_0 is the unit normalizing factor. This is expressed in dB as

$$\frac{10\log A(l,f)}{A_0} = k.10\log l + l.10\log\alpha(f) \tag{36}$$

The absorption coefficient is empirically given by Thorps equation which is valid for lower frequencies [18]

$$10\log\alpha(f) = 0.11\frac{f^2}{1+f^2} + 44\frac{f^2}{4100+f^2} + 2.75.10^{-4} f^2 + 0.003 \tag{37}$$

As the frequency increases the absorption coefficient (dB/Km) increases exponentially, as shown in Fig 3.

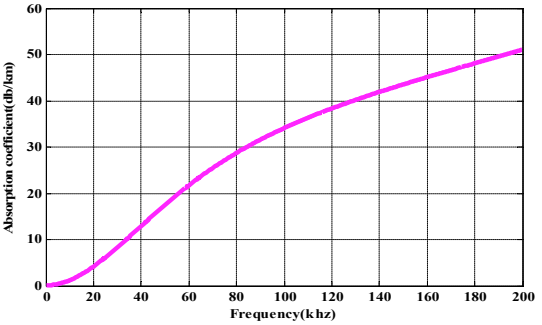


Fig 3. Frequency vs. Absorption Coefficients (dB/Km)

There are four different types of noise in UWAC. The thermal noise $N_{th,}$ which is considered as AWGN noise for any communication system, noise caused due to the motion of waves N_w Noise caused by the ship traffic N_s and the noise due to turbulence N_t. The corresponding equations are given by [19]

$$N_{th} = 10^{\frac{((-15+20\log(f))}{10}} N_{th_{off}} \tag{38}$$

$$N_w = 10^{\frac{(50+(7.5(\sqrt{w}))+(20log10(f))-(40log10(f+0.04))}{10}} N_{w_{off}} \tag{39}$$

$$N_s = 10^{\frac{((40+(20(s-0.5))+(26log10(f))-(60log10(f+0.03)))}{10}} N_{s_{off}} \tag{40}$$

$$N_t = 10^{\frac{((17-30log10(f))}{10}} N_{t_{off}} \tag{41}$$

Total Noise $$N = N_{th} + N_w + N_s + N_t \tag{42}$$

As in seawater, there is ambient noise; the signal gets attenuated by the unwanted noise, increasing the bit error rate [20]. Apart from the ambient noise, multipath is also present in the underwater channel that causes ISI. To overcome this effect, Equalization techniques are used, which are clearly explained in Section 4.

4. Equalization Techniques

Because of the time variance property of the underwater channel, multipath exists in UWAC, which results in ISI. This destructs the transmitted data, causing errors at the receiver. ISI has been identified as the major drawback in high-speed UWAC. Generally, in Single input single-output (SISO) cases, a great fraction of transmitting power is focussed away from the receiver, which results in severe degradation of the performance [21]. This technique is generally used to compensate the ISI, which is created by the multipath propagation in UWAC [22]. To combat the channel variations in UWAC, efficient equalization methods need to be used [23]. The inverse of the channel

frequency response is applied at the received signal in ZF Equalizer. This is done to repair the transmitted signal at the receiver. The name Zero Forcing relates to taking down the ISI to zero in a noiseless case. This will be beneficial when ISI is noteworthy compared to noise. Along with ISI, the additive noise effects are also reduced using MMSE Equalizer [24]. The flow chart shown in Fig 4 represents the flow in which equalization techniques are adapted for UWAC.

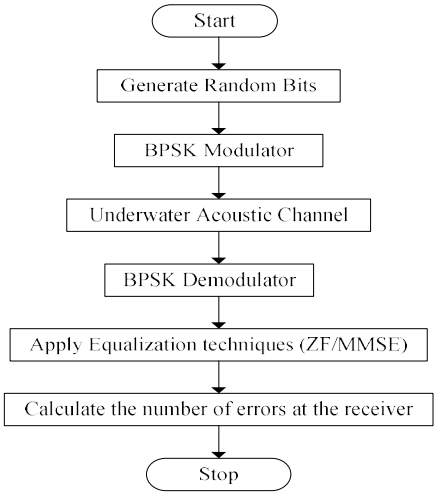


Fig 4. Flowchart for Equalization Technique

The flow chart shows that the equalization technique is applied only after the data is demodulated. Let us consider in detail the two different equalizers used in this paper.

4.1. ZF Equalizer

In this ZF Equalizer, the inverse of the frequency response of the channel is calculated. The zero forcing equalizer $Z(f)$ is given by

$$Z(f) = \frac{1}{C(f)} \tag{43}$$

Where *C(f)* is the frequency response of the underwater channel. In this case, it should fulfil the criteria

$$C(f)Z(f) = 1 \tag{44}$$

The received data, in any case, is given by

$$Y = HX + N \tag{45}$$

Here H is the underwater channel with multipath propagation and N is the ambient noise mentioned in Section 3. X and Y are the transmitted and received signals respectively. After applying ZF equalizer, the estimate of X is expressed as

$$\hat{X} = H^{-1}(HX + N) \tag{46}$$

$$\hat{X} = H^{-1}HX + H^{-1}N \tag{47}$$

$$\hat{X} = IX + H^{-1}N \tag{48}$$

As the transmitted data is estimated by applying the inverse matric, the outcome comes at the expense of increased noise. To overcome this, MMSE Equalizer came into existence which is explained further**.**

4.2. MMSE Equalizer

MMSE Equalizer follows an approximation technique by which it reduces the mean square error; a common compute to know the quality of the estimator [25]. This technique generally minimizes the ISI at the output and the

total power of the noise. MMSE methodology efforts to find a coefficient W which minimizes the criterion. According to the orthogonality principle [26]

$$E\left\{\left[W_{y-x}\right]\left[W_{y-x}\right]^{H}\right\} \tag{49}$$

The above equation should fulfill the criteria $WH = I$

$$W_{MMSE} = \left(H^{*}H + NI\right)^{-1} H^{*} \tag{50}$$

H^{*} is the hermitian. This receiver technique reduces both interference and noise components, but as far as ZF is concerned, it holds back only the interference [27]. Linear MMSE estimators are widespread because of its simplicity and adaptable nature.

5. Simulation Results

When a comparison of the modulation schemes with the probability of error equations as given in section 2 is made, the BER curve is as shown in Fig 5. A few similarities in the curves are that BPSK, QPSK, i.e., 4-PSK and 4-QAM have the same probability of error. In its most known form, QPSK is identical to 4-QAM. That is, the same constellation can be referred to by different names. So there is no difference in terms of error calculation.

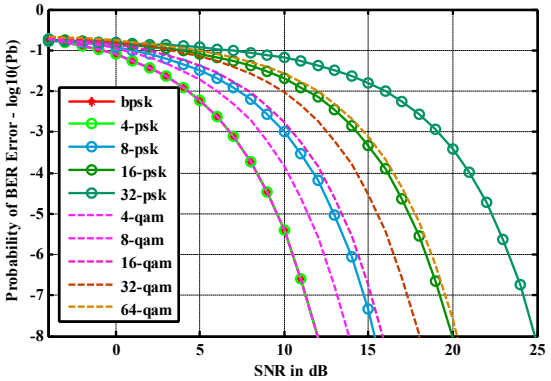
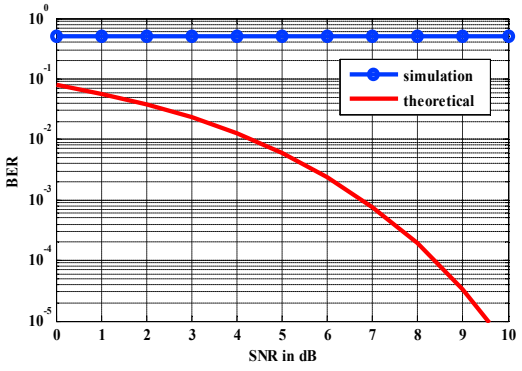


Fig 5. BER vs. SNR for all modulation techniques

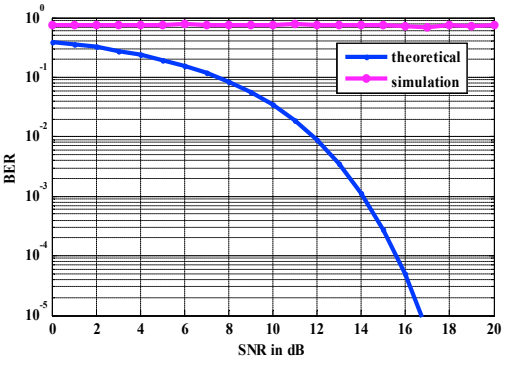
Let us consider the individual modulation scheme and compare the theoretical error performance to the practical error caused after simulating the underwater channel consisting of the ambient noise and scattering effects. The measure of the performance is done using the BER. A BER is a rate at which errors are occurring in the transmission system. It is also expressed as the ratio of No. of Errors to the stated number of bits. It is defined as

$$BER = \frac{No\ of\ Errors}{Total\ No\ of\ Bits\ Sent} \tag{51}$$

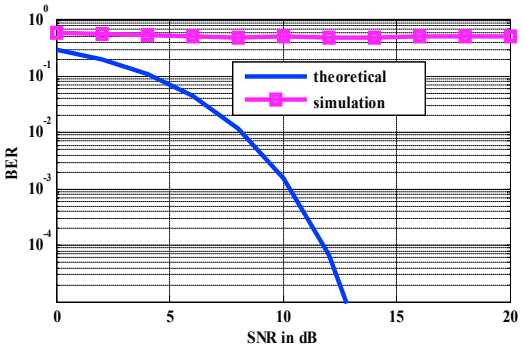
The following Fig 6(a), (b) and (c) gives the BER performance of BPSK, 4-PSK, and 4-QAM, respectively.



(a)



(b)



(c)

Fig 6. (a) BER vs. SNR for BPSK Modulation, (b) BER vs.SNR for 4-PSK Modulation. (c) BER vs. SNR for 4-QAM Modulation

In the above-shown graphs, the different modulation schemes BPSK, 4-PSK, and 4-QAM are individually considered. But the practical error because of the presence of multipath and the ambient noise with all the scattering effects is never matching the performance of the probability of error. To overcome the effects, equalizers are used at the receiver to further improve the performance of the UWAC system. Two equalizers used in this paper are ZF Equalizer and MMSE Equalizer, which is discussed in Section 4.

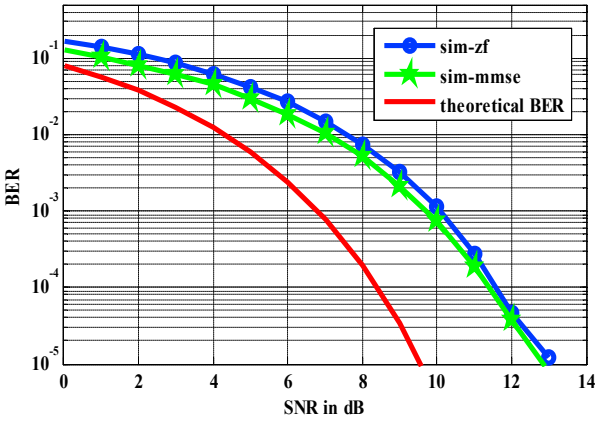


Fig 7 BER vs. SNR for Equalization Techniques

From Fig 7, it is observed that as the equalization techniques are applied to the practical UWAC channel, the errors are further decreased. This improves the performance of the system practically. At lower SNR, the noise is also amplified in case of ZF Equalizer, and the MMSE Equalizer reduces all these noise effects as well. But at very high SNR, both ZF and MMSE equalizers perform in an equal manner.

6. Conclusions and Future Scope

In this paper, studies have been made on different modulation techniques for UWAC. It is observed that the scattering effects and the multipath with ambient noise in the underwater environment impact the performance of the UWAC system. To overcome these effects, ZF and MMSE equalization techniques are used to reduce ISI. However, ZF comes at the expense of the enhancement of the noise, but MMSE reduces all the noise levels. Equalization Techniques that can be adapted for higher distance and higher signal power transmission are to be investigated further. Multi-input multi-output (MIMO) can be considered in UWAC to overcome the bandwidth limitation of the undersea channel. There is still a very high scope for this paper as the major part of the ocean bottom is still unexplored. Also, marine archaeology needs to find its applications of modulation techniques in UWAC.

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